

INDUSTRIAL PRODUCT DATASHEET

# reServer Industrial J501- Carrier board for Jetson AGX Orin

Industrial carrier board for Jetson AGX Orin with up to 275 TOPS, advanced vision and connectivity, and -25°C to +60°C operation for AI robotics, vision AI, and edge generative AI.

SKU

102991854

UPDATED

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PRICE

\$379.00(Excl. VAT)



# Overview



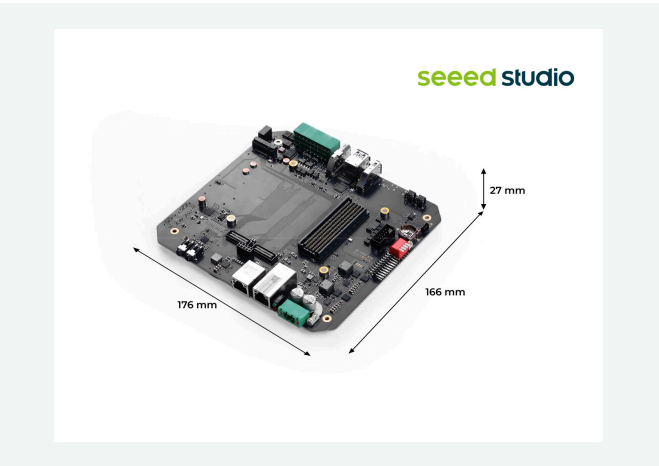
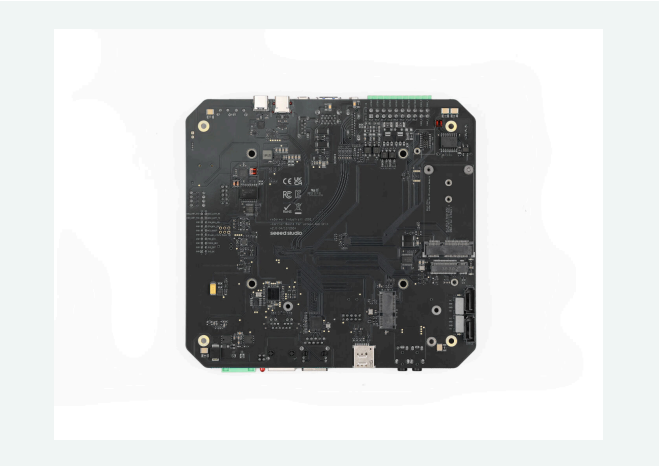
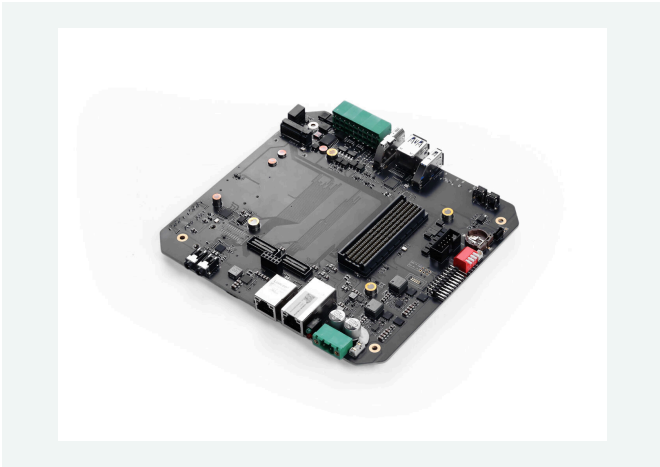
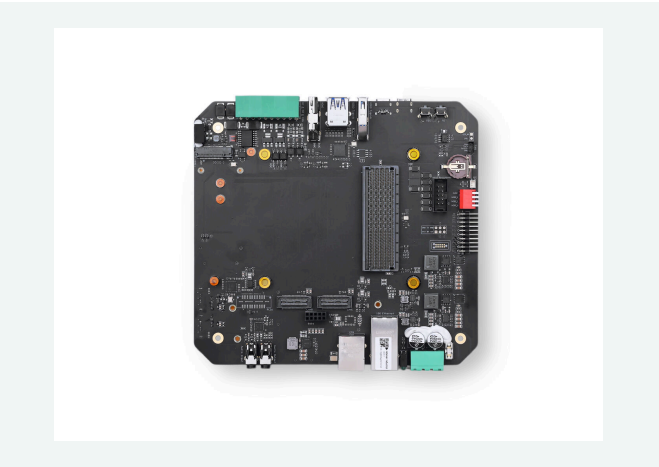
The reServer Industrial J501-Carrier board for Jetson AGX Orin delivers up to 275 TOPS with configurable 15–60W power, 8K60 and 3x4K60 video decoding, 16 lanes of MIPI CSI-2, and optional GMSL support for up to 8 cameras. It provides extensive connectivity including Gigabit Ethernet, 10G Ethernet, USB 3.1, M.2 Key E/B/M, CAN, RS232/422/485, operates across -25°C to +60°C with 12–36V DC input, and supports Jetpack 6 for AI robotics, vision AI, and generative AI at the edge.

# Key Features

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- **Build the most powerful embedded AI platform:** Compatible with the Jetson AGX Orin module, offering up to 275 TOPS—8 times more than Jetson AGX Xavier. Power is configurable between 15W and 60W.
- **Advanced Vision AI Capabilities:** Supports 8K60 and 3x4K60 video decoding. Features 16 lanes of MIPI CSI-2, with an optional GMSL extension board supporting up to 8 GMSL cameras.
- **Extensive Connectivity:** including Gigabit Ethernet, 10G Ethernet, USB 3.1, M.2 Key E, Key B, and Key M. CAN and RS232/422/485 are supported for connecting to industrial devices and sensors.
- **Wide Operating Range:** Operates reliably across a temperature range of -25°C to +60°C and an input voltage range of 12-36V DC.
- **Support Jetpack 6:** BSP ready for developing your custom system image, leverage Jetson Platform Service for end to end workflow.

# Product Gallery

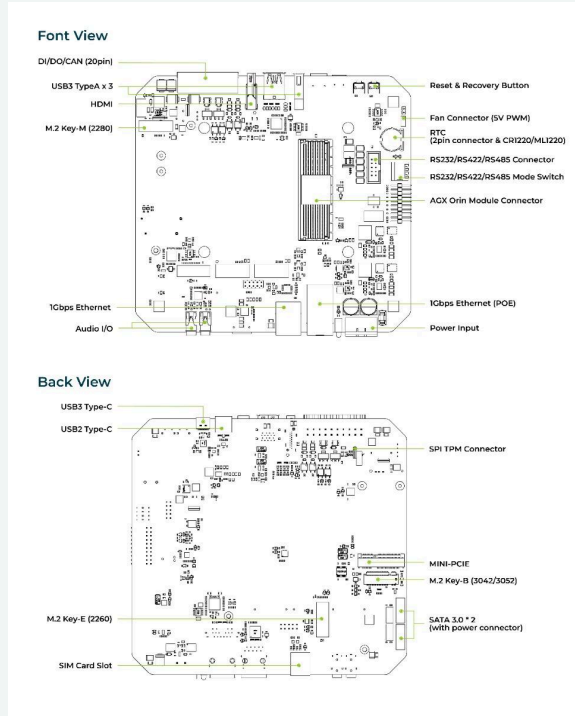


## Specification

|                                        |                                |                                |
|----------------------------------------|--------------------------------|--------------------------------|
| Compatible with Jetson AGX Orin Module |                                |                                |
| AI Performance                         | Jetson AGX Orin 64GB, 275 TOPS | Jetson AGX Orin 32GB, 200 TOPS |

|                    |                                                                             |                                                                              |                                                                                                |
|--------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| GPU                | NVIDIA Ampere architecture with 2048 NVIDIA® CUDA®cores and 64 Tensor cores |                                                                              | 1792-core NVIDIA Ampere architecture GPU with 56 Tensor Cores                                  |
| CPU                | 12-core Arm Cortex-A78AE v8.2 64-bit CPU<br>3MB L2 + 6MB L3                 |                                                                              | 8-core Arm® Cortex®-A78AE v8.2 64-bit CPU<br>2MB L2 + 4MB L3                                   |
| DL Accelerator     | 2x NVDLA v2.0                                                               |                                                                              |                                                                                                |
| Vision Accelerator | PVA v2.0                                                                    |                                                                              |                                                                                                |
| Memory             | 64GB 256-bit LPDDR5 204.8GB/s                                               |                                                                              | 32GB 256-bit LPDDR5 204.8GB/s                                                                  |
| Storage            | 64GB eMMC 5.1                                                               |                                                                              |                                                                                                |
| Video Encode       | 2x 4K60   4x 4K30   8x 1080p60   16x 1080p30 (H.265)                        |                                                                              | 1x 4K60 (H.265)   3x 4K30 (H.265)   6x 1080p60 (H.265)   12x 1080p30 (H.265)                   |
| Video Decode       | 1x 8K30   3x 4K60   6x 4K30   12x 1080p60   24x 1080p30 (H.265)             |                                                                              | 1x 8K30 (H.265)   2x 4K60 (H.265)   4x 4K30 (H.265)   9x 1080p60 (H.265)   18x 1080p30 (H.265) |
| Power              | 15W-60W                                                                     |                                                                              |                                                                                                |
| Carrier Board      |                                                                             |                                                                              |                                                                                                |
| Storage            |                                                                             |                                                                              | 1x M.2 Key M (PCIe 4.0)                                                                        |
| Networking         | Ethernet                                                                    | 1 x LAN0 RJ45 GbE(10/100/1000Mbps)<br>1 x LAN1 RJ45 10GbE(10000Mbps)         |                                                                                                |
|                    | M.2 Key E                                                                   | 1x M.2 Key E for WiFi/Bluetooth module                                       |                                                                                                |
|                    | M.2 Key B                                                                   | 1x M.2 Key B (3042/3052) support 4G/5G (Module not included)                 |                                                                                                |
|                    | Mini PCIE                                                                   | 1x Mini PCIE for LoRaWAN®/4G/5G Wireless (Module not included)               |                                                                                                |
| I/O                | USB                                                                         | 3x USB3.1;<br>1x USB3.1 Type C(Host mode),<br>1x USB2.0 Type C(Device mode), |                                                                                                |
|                    | DI/DO                                                                       | 4x DI,4x DO,3x GND_DI,2x GND_DO,1x GND_ISO,1x CAN<br>1x RS232/RS422/RS485    |                                                                                                |
|                    | Display                                                                     | 1x HDMI 2.1 Type A 7680x4320                                                 |                                                                                                |
|                    | SATA                                                                        | 2x SATA III 6.0Gbps) at 30 Hz                                                |                                                                                                |
|                    | SIM                                                                         | 1x Nano SIM card slot                                                        |                                                                                                |
|                    | Button                                                                      | Reset Button, Recovery Button                                                |                                                                                                |
|                    | Fan                                                                         | 1x Fan connectors (5V PWM)                                                   |                                                                                                |
|                    | TPM                                                                         | 1x TPM 2.0 connector (Module not included)                                   |                                                                                                |
|                    | RTC                                                                         | 1x RTC socket (CR1220 included)<br>1x RTC 2-pin                              |                                                                                                |
| Expansion          | Camera (For GMSL Extension Board)                                           | 2x Expansion connector (8 lanes for each connector)                          |                                                                                                |
|                    | PCIE                                                                        | 1x PCIE                                                                      |                                                                                                |
| Power              | Power Supply                                                                | DC 12V-36V Terminal block 2 pin                                              |                                                                                                |
|                    | Power Adapter                                                               | 24V/5A Power Adapter(Without power cord)                                     |                                                                                                |
| Mechanical         | Dimensions (W x D)                                                          | 176 x 163mm (Module not included)                                            |                                                                                                |
|                    | Operating Temperature                                                       | -20~60°C                                                                     |                                                                                                |
|                    | Weight                                                                      | 225g (Module not included)                                                   |                                                                                                |

# Hardware Overview



## Applications

- **AI Robot Development:** With AGX Orin module, this most powerful embedded AI system lets you create advanced robotics and edge AI applications for manufacturing, logistics, and healthcare. It's the most powerful embedded system, pushing the limits of AI robotics. The Isaac robotics platform offers NVIDIA-accelerated systems, libraries, and generative AI models to enhance AI perception, manipulation, and simulation in robotics.
- **Vision AI:** Jetson AGX Orin supports real-time video analytics with up to 23 1080P streams simultaneously. NVIDIA Metropolis, an application framework that combines visual data and AI, equips developers with the tools and ecosystem to enhance operational efficiency and safety across diverse industries.
- **Generative AI at the Edge:** Jetson AGX Orin offers unmatched AI compute power, large unified memory, and comprehensive software stacks, making it the ideal platform for energy-efficient generative AI applications. It supports fast inference for any generative AI model based on the transformer architecture and leads in edge performance according to MLPerf benchmarks. Quickly deploy generative AI models on text generation, multimodal models, image generation via jetson-examples.

## Part List

|                                                                 |    |
|-----------------------------------------------------------------|----|
| reServer Industrial J501 Carrier Board for Jetson AGX Orin      | x1 |
| Power Adapter AC Power Brick 120W<br>(Power cable not included) | x1 |
| Terminal Connector 20pin                                        | x1 |
| 2-Pin Terminal Block Power Connector                            | x1 |

## Compliance & Logistics



|          |            |
|----------|------------|
| HSCODE   | 8543909000 |
| USHSCODE | 8517180050 |
| UPC      |            |
| EUHSCODE | 8543709099 |
| COO      | CHINA      |

### FCC Requirement

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

(1) this device may not cause harmful interference, and

(2) this device must accept any interference received, including interference that may cause undesired operation.

15.21

This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This equipment generates, uses, and can radiate radio frequency energy.

If not installed and used in accordance with the instructions, it may cause harmful interference to radio communications.

However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

# Resource Links

- <https://www.seeedstudio.com/reServer-Industrial-J501-Carrier-board-for-Jetson-AGX-Orin-p-5950.html>
- [Datasheet](#)
- [Schematic](#)
- [3D File](#)
- [Seeed Nvidia Jetson Product Catalog](#)
- [Nvidia Jetson Comparison](#)
- [Nvidia Jetson Product Comparison](#)
- [Seeed Nvidia Jetson Success Cases](#)
- [Seeed Jetson One Pager](#)
- [Flash BSP for Jetson](#)

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